ME 274: Basic Mechanics II – Spring 2026

School of Mechanical Engineering
Purdue University, West Lafayette and Indianapolis

Course description Particle kinematics (both 2D and 3D motion, including moving reference frame descriptions for each). Particle kinetics including Newton's laws, work/energy and impulse/momentum. Planar kinetics of rigid bodies. Kinetics for planar motion of rigid bodies including equations of motion and principles of energy and momentum. Introduction to linear vibrations with emphasis on single-degree-of-freedom systems.

Course schedule A topical schedule for the course is attached at the end of this syllabus.

Course instructors

Course TAs

Course Textbook

The required course lecturebook: *Dynamics: A Lecturebook, 3rd Edition,* can be purchased from the University Bookstore.

Course website

https://www.purdue.edu/freeform/me274/

The course website will be used for both threaded discussions and for the delivery of all course material. Example problem solutions from the lecturebook are posted as multimedia video/audio files on the course website. Use these solutions to help assess your ability to work the homework problems for that day. Homework video solutions will be made available on the website following the date of submission. Animations and simulations are presented on the website to assist you in visualizing motion and other concepts related to the topics of the course. All information related to homeworks and exams is available on the website. ***Please note that there will be NO Brightspace presence in the course since all course material will be delivered to you through the course website.***

Class meetings

Your attendance is expected for all meetings during the semester. If you are unable to attend a class meeting, you are expected to contact your instructor prior to the start of the class period.

Homework

There will be a homework assignment for every regular class period of the semester. Two problems will be included in each assignment. The problem statements will be posted on the course website on the date of the assignment. The homework is to be submitted on Gradescope by 11:59PM on the day of the next regular class period. Please see the pages on "*Homework guidelines*" and "*Homework format*" near the end of this document.

Course grading

Your course grade will be based on a straight grading scale:

97-100% A+
93-97% A
90-93% A-
87-90% B+
83-87% B
80-83% B-

77-80% C+
73-77% C
70-73% C-
67-70% D+
63-67% D
60-63% D
< 60% F.

The breakdown percentages for the components of your course grade are the following:

1. *Homework and quizzes*: 25%
   Two homework problems are assigned per lecture. Completed homework assignments are to the submitted on Gradescope by 11:59PM of the due date. *Late homework will not be accepted.* A sample format is attached. Your work needs to be presented with a logical thought process and in a neat, easy-to-read style. Failure to do so can result in a loss of points in your homework grade. We will collect 39 homework assignments during the semester. At the end of the semester, your course homework score will be based on 36 homework assignments. That is, *you are able to miss submissions for 3 homework sets without penalty to your grade*.

2. *Midterm and final exams*: 75%
   You will be given two, one-hour midterm term exams during the term and a final exam during the scheduled University final exam period. At the end of the term, the average of the two midterm exams will be compared against your final exam score. The higher of these two will be given a 50% weighting and the lower of these two will be given a 25% weighting in the computation of your course average from which your course grade is determined.

3. *Bonus points:* Your instructor may also announce additional homework/quiz bonus points for your section throughout the semester. *Your total combined HW/Quizzes/Bonus points will not exceed 25% of your final grade*.

Tutorial assistance

The discussion thread of the course website will be open 24/7 throughout the course. We will be closely monitoring this discussion thread, and you can expect a quick turn-around on your questions during normal daytime hours. During the remaining time, we will continue to monitor the discussion and provide assistance when needed. In addition, the course instructors and TAs will be providing interactive tutorial sessions/office hours. Your instructor will provide you with a schedule of tutorial/office hours during the first week of class. You are free to attend any of the office hours/tutorial sessions, as you are not restricted to just the office hours of your instructor.

Collaboration

Faculty and students working together can promote a fair and positive academic environment. All students are expected to conduct themselves in an ethical manner. Students are permitted to discuss homework assignments together, but should do their own work when preparing a problem solution). Any of the following is strictly prohibited in this course: copying from solution manuals, online resources (e.g., Chegg), or another student's work; using artificial-intelligence or machine-learning tools (e.g., ChatGPT); and posting course problems to online cheating services (e.g., Chegg). Please note that AI tools frequently get things wrong and cannot be relied upon for factual accuracy. Submitting solutions copied from the output of AI tools shortcuts your learning process and provides you with a false sense of learning. Exams are to be completed without unauthorized assistance.

Academic Integrity

Your instructor is a strong believer that the mission of the University can only be fulfilled in an environment of high academic integrity. Students are encouraged to review the University's policies on academic integrity, which is located at:
https://www.purdue.edu/odos/academic-integrity/
In addition, the Purdue University Honor Pledge is located at:
https://www.purdue.edu/provost/teachinglearning/honor-pledge.html
Suspected cases of academic integrity violations will be reported to the Dean of Students.

Questions?

Please see the following page entitled "*Some common questions at the start of the semester.*", and/or contact your instructor.

Emergency Preparedness

Though Purdue University is continuously preparing for natural disasters and human-caused incidents with the ultimate goal of maintaining a safe and secure campus, emergency preparedness is a personal responsibility. Please review the following items:

- To report an emergency, call 911.
- To obtain updates regarding an ongoing emergency, and to sign up for Purdue Alert text messages, view *www.purdue.edu/ea*
- There are nearly 300 Emergency Telephones outdoors across campus and in parking garages that connect directly to the Purdue Police Department (PUPD). If you ever feel threatened or need help, push the button, and you will be connected immediately.
- If we hear a fire alarm, we will immediately suspend class, evacuate the building, and proceed outdoors away from the building. Do not use the elevator.
- Emergency assembly area for ME building occupants:
  - Primary: Purdue Mall area outside MSEE.
  - Secondary (in case of inclement weather): Atrium of the MSEE building, located in the center of the building's first floor.
- If we are notified of a Shelter in Place requirement for a *tornado* warning, we will suspend class and shelter in the lowest level of the ME building away from windows and doors.

- If we are notified of a Shelter in Place requirement for a *hazardous materials release*, or a *civil disturbance*, including a shooting or other use of weapons, we will suspend class and shelter in our classroom, shutting any open doors or windows, locking or securing the door, and turning off the lights. If you are directed to shelter in place, but you are unaware of the specific reason, proceed to the lowest level of the building but continue to seek additional information by all possible means to determine the type of incident. Once you have determined the type of emergency, follow the below chart:

Emergency	***Emergency Assembly Area (EAA)***
Weather-Related / Tornado Warning	Basement corridors, basement offices, basement restrooms or the lowest level of the building (stay away from windows and doors)
Hazardous Materials (HAZMAT) Release	Remain or find an unaffected office or work area and close windows and doors
Civil Disturbance / Active Shooter	Seek a safe location, preferably a room without windows that can be locked or secured by barriers

Please review the Emergency Preparedness website for additional information:
http://www.purdue.edu/ehps/emergency_preparedness/index.html

Diversity & Inclusion Purdue University is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach his or her own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life. Purdue's nondiscrimination policy can be found at: http://www.purdue.edu/purdue/ea_eou_statement.html.

Mental health Mental Health If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try Therapy Assistance Online (TAO), a web and app-based mental health resource available courtesy of Purdue Counseling and Psychological Services (CAPS). TAO is available to all students at any time by creating an account on the TAO Connect website, or downloading the app from the App Store or Google Play. It offers free, confidential well-being resources through a self-guided program informed by psychotherapy research and strategies that may aid in overcoming anxiety, depression and other concerns. It provides accessible and effective resources including short videos, brief exercises, and self-reflection tools.

If you need support and information about options and resources, please contact or see the Office of the Dean of Students. Call 765-494-1747. Hours of operation are M-F, 8 a.m.- 5 p.m.

If you find yourself struggling to find a healthy balance between

academics, social life, stress, etc., sign up for free one-on-one virtual or in-person sessions in West Lafayette with a Purdue Wellness Coach at RecWell. Student coaches can help you navigate through barriers and challenges toward your goals throughout the semester. Sign up is free and can be done on BoilerConnect. Students in Indianapolis will find support services curated on the Vice Provost for Student Life website.

If you're struggling and need mental health services: Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at 765-494-6995 or http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or by going to the CAPS offices in West Lafayette or Indianapolis.

Accessibility

Purdue University strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on disability, you are welcome to let me know so that we can discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.